23CY44/23CI44

RAMAIAH
Institute of Technology

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(Approved by AICTE, New Delhi & Govt. of Karnataka)
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SEMESTER END EXAMINATIONS – JULY / AUGUST 2025

Program	:	B.E. – CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)	Semester	:	IV
Course Name	:	Automata Theory and Compiler Design	Max. Marks	:	100
Course Code	:	23CY44 / 23CI44	Duration	:	3 Hrs.

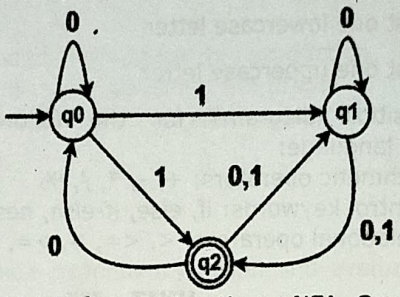
Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Design DFAs to accept the following languages. CO4 (09)
   i. String beginning with 'bba' over Σ= {a,b}.
   ii. String not ending with 01 over Σ= {0,1}.
   iii. String starting with 'aa' and ending with 'bb'.

   b) Convert the given NFA to DFA. CO4 (07)



   c) Describe the components for constructing a NFA. Construct an NFA with Σ = {0, 1} that accepts all strings that begin with a and end with ab. CO4 (04)

2. a) Explain the analysis and synthesis phase of the compiler by translating the assignment statement. Assume all variables are integers. CO1 (08)

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Res=a*a+2*a*b+b*b;
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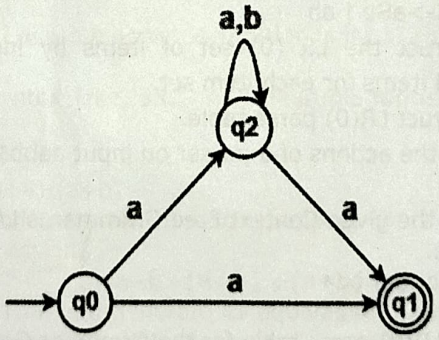
   b) Design an NFA with Σ = {0, 1} which accepts the language consists of all the strings containing substring 1010. CO4 (05)

   c) Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA). Construct the Transition Table. CO4 (07)



UNIT - II

3. a) How does a lexical analyzer employing a two-buffer scheme with sentinel characters process the lexeme a<=b in a source line, and what benefits does this method provide in efficiently handling look-ahead scenarios? CO2 (06)

b) Describe the language accepted by the regular expression: CO2 (06)

i) (a|b)aab*

ii) (ab|ba)*

iii) (a|b)*b(a|b)

c) Construct the pushdown automata for accepting following languages: CO2 (08)

i) $\{a^n b^n \mid n > 0\}$

ii) $\{a^n b^{2n} \mid n > 0\}$

4. a) How does the separation of lexical analysis from parsing influence the clarity, modularity, and maintainability of a compiler, and how do the roles of tokens, patterns, and lexemes contribute to this design choice? CO2 (06)

b) i) Write a regular expression to validate a simple email address format. CO2 (05)

ii) Construct a regular expression that enforces the following password rules:

- At least one digit
- At least one lowercase letter
- At least one uppercase letter

c) Design transition diagrams for the following components of a programming language: CO2 (09)

(i) Arithmetic operators: +, -, *, /, %

(ii) Control keywords: if, else, if-else, nested if, and nested for

(iii) Relational operators: <, <=, >, >=, ==, !=

UNIT - III

5. a) Consider the following Context Free Grammar: CO3 (10)

S->SA |0 |ε

A->aS1 | a

i. Compute the FIRST and FOLLOW sets for each non-terminal symbol.

ii. Construct the parsing table for a non-recursive predictive parser for the above grammar.

iii. Is the grammar LL (1)? Justify.

b) Consider the grammar CO3 (10)

G:- S->aSb | ab

i) Construct the LR (0) set of items by indicating the Kernel and Non-kernel items for each item set.

ii) Construct LR(0) parse table.

iii) Show the actions of a parser on input aabb$

6. a) Prove that the given Context Free Grammar is LALR(1) by generating the parse table. CO3 (10)

S->Aa| bAc| dc| bda

A->d

b) Construct LR(0) parse table for the Grammar Given CO3 (10)

G: S-> L

L-> (L)L | a | ε

Do parsing for the input: (a)$

UNIT- IV

7. a) Design a syntax-directed definition (SDD) using only synthesized attributes to evaluate arithmetic expressions involving digits, the operators +, *, and parentheses for the following grammar. CO4 (10)

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid digit$

i) Include semantic rules to compute the val attribute for each nonterminal.

ii) Use digit.lexval to refer to the integer value of a digit, as returned by the lexical analyzer.

Then, using your SDD, construct the annotated parse tree for the input expression: 3 * 5 + 4.

b) Obtain a Turing machine to accept the language. CO4 (10)

$L = \{0^n1^n \mid n >= 1\}$

8. a) Given the production and semantic rule: CO4 (10)

$E \rightarrow E_1 + T$ with $E.val = E_1.val + T.val$ and the following values from a parse tree:

$E_1.val = 10$

$T.val = 5$

(i) Draw the dependency graph for this production.

(ii) Compute the value of E.val using the graph.

(iii) Extend the production to a larger expression:

$E \rightarrow T$

$T \rightarrow T_1 + F$

$T \rightarrow F$

$F \rightarrow digit$

Construct a complete dependency graph and evaluate the final value of E.val.

b) Describe the language of a Turing Machine. Specify the Turing Machine and halting problem. CO4 (10)

UNIT - V

9. a) Translate the following control flow statement with the help of semantic actions of one pass code generation. CO5 (10)

if(a<b) a=1; else a=0;

b) Translate the following array reference assignment statements with the help of SDT. CO5 (10)

i) x=a[i]+b[j]

ii) c=a+b[i][j] assume array declaration is int b[2][2]

10. a) Construct DAG, Syntax Tree, 3AC and Quadruple representation for CO5 (10)

i. x=a+b*-c+b*-c

ii. x=a+b+b+a+a*b

iii. x=(a+b)*c+(d+a)-(a+b)

b) Construct DAG and generate its corresponding three address code for the following expression. CO5 (10)

a=a+b+(a+b) +(a+b)

Draw the Syntax Tree and generate its equivalent three address code for the same expression. Does both generates the same three address code sequence? If not, state the reason.

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